Artificial Intelligence and Machine Learning

1. Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence. These tasks include reasoning, problem-solving, language understanding, and image recognition.

Examples of AI applications include virtual assistants, recommendation systems, autonomous vehicles, and chatbots.

2. Machine Learning

Machine Learning (ML) is a subset of AI that allows computers to learn patterns from data and make predictions without being explicitly programmed for every task.

There are three main types of Machine Learning:

- **Supervised Learning:** Learns from labeled data. Examples include Linear Regression and Decision Trees.
- **Unsupervised Learning:** Finds patterns in unlabeled data. Examples include K-Means Clustering and PCA.
- **Reinforcement Learning:** Learns through rewards and penalties.

3. Deep Learning

Deep Learning is a subset of Machine Learning that uses neural networks with multiple layers. It is useful for solving complex problems involving images, audio, and text.

Convolutional Neural Networks (CNNs) are commonly used for image classification. Recurrent Neural Networks (RNNs) can process sequential data, while Transformers are widely used in modern language models.

4. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) combines information retrieval with a large language model.

A typical RAG workflow includes:

1. Loading documents such as PDF files.
2. Splitting documents into smaller text chunks.
3. Converting chunks into vector embeddings.
4. Storing embeddings in a vector database.
5. Retrieving relevant chunks when a user asks a question.
6. Sending retrieved context to a language model to generate an answer.

5. Embeddings and Vector Databases

Embeddings represent text as numerical vectors. They help applications identify semantically similar content. Vector databases store embeddings and enable similarity searches. Examples include ChromaDB, FAISS, and Pinecone.

6. React and AI Applications

React is a JavaScript library used to build interactive user interfaces. In an AI document-question-answering application, React can provide a file upload component, a question input box, and an area to display answers.

The frontend can communicate with a backend API that extracts PDF text, retrieves relevant information, and generates answers using a language model.

7. Example Questions

1. What is Artificial Intelligence?
2. What are the three types of Machine Learning?
3. What is the difference between CNNs and RNNs?
4. What does RAG stand for?
5. Why are embeddings used in a RAG pipeline?
6. What role does React play in an AI application?

Conclusion

AI, Machine Learning, Deep Learning, and RAG are important technologies in modern software development. A React-based document-question-answering application can combine a user-friendly frontend with PDF processing, semantic search, and language model capabilities.